B. KiaKio and Squared Numbers

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: brute force, implementation | https://codeforces.com/problemset/problem/2269/B

Kia and Kio spent the summer at the port of Mehragan, where n lighthouses stand on the cliffs facing the dark sea.

The lighthouses of Mehragan do not give light. Every night a **number** is written in fire on each of them, and the sailors read their way from those numbers.

The law of the lighthouses is this: if a lighthouse shows x tonight, then tomorrow night it shows the **sum of the squares of the decimal digits** of x.

For example, a lighthouse showing 23 will show $2^2+3^2=13$ tomorrow, then $1^2+3^2=10$, and then 1.

On night 0 of the season, lighthouse i shows the number a_i. From that night on, the law is applied once every night, forever.

Kio calls two lighthouses i and j **in tune** if there exists a night after which, **forever**, both of them show exactly the same number on every single night.

Kia asks: how many pairs (i, j) with i lt j are in tune?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 1000$). The description of the test cases follows.

Each test case consists of two lines.

The first line of each test case contains a single integer n ($1 \le n \le 1000$) — the number of lighthouses.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the number shown by each lighthouse on night 0.

It is guaranteed that the sum of n over all test cases does not exceed 1000.

Output

For each test case, print a single integer — the number of pairs (i, j) with i lt j such that lighthouses i and j are in tune.

Example

Input

```
4
5
7 4 16 4 2
4
1 7 10 100
3
4 16 37
3
2 20 4
```

Output

```
1
6
0
1
```

Note

In the first test case:

- Lighthouse 1 starts at 7: 7 → 49 → 97 → 130 → 10 → 1, and it stays at 1 forever.
- Lighthouses 2 and 4 both start at 4, so they show the same number on every night.
- Lighthouse 3 starts at 16 and lighthouse 5 starts at 2; each of them is at a different point of the cycle Kio found, and never matches anybody.

So the only pair in tune is (2, 4), and the answer is 1.

In the second test case, every lighthouse sooner or later reaches 1 and stays there, so all of them are pairwise in tune, which gives 6 pairs.

In the fourth test case, on night 1, lighthouse 2 shows $2^2+0^2 = 4$, and lighthouse 1 shows $2^2 = 4$. From night 1 on, they are identical forever. Lighthouse 3 starts at 4, so it is one night ahead of both of them and is in tune with neither.